JORGE

ANAIS VILCHEZ

CONTACT

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EXPERIENCE

Research **2022 - Present**

Characterization of the Sagittarius dwarf spheroidal galaxy at low Galactic latitudes | *Universidad de Antofagasta*

Principal investigator of the project funded by the ANID National Doctoral Scholarship (2024-2026). Detailed characterization of the structure, stellar content, and dynamics of the Sagittarius dwarf galaxy in low Galactic latitude regions, combining photometric and spectroscopic data from multiple surveys.

Instructor **2023 - Present**

Duoc UC, School of Informatics and Telecommunications | *Santiago, Chile*

Teaching Machine Learning and Deep Learning courses for computer engineering students.

Early Career Visitor Programme **04/2025 - 10/2025**

Characterization of new Sagittarius Dwarf Galaxy Clusters | *ESO, Chile*

Researcher in the project led by Dr. Elisa R. Garro combining deep HST photometry (WFC3, F555W and F814W) and FLAMES+UVES spectroscopy (98 hrs, PI: Garro) to precisely determine ages and chemical compositions of Sagittarius dwarf galaxy globular clusters. Goals include resolving the age-metallicity relation, testing formation scenarios during Sgr-Milky Way interaction, and comparing with theoretical models to reconstruct Sgr's evolutionary history.

Research **09/2024 - Present**

Variable stars in the Sagittarius dwarf spheroidal galaxy | *NOIRLab*

Collaboration with NOIRlab researcher Kathy Vivas to detect and classify variable stars (focus on δ Scuti and RR Lyrae stars) in the central regions of Sagittarius galaxy using multiepoch DECam data.

Data Scientist **08/2021 - 04/2022**

Grupo Kaufmann AWTO | *Chile and Brasil*

Design and implement machine learning models, perform exploratory data analysis, communicate insights and recommendations to stakeholders.

Research **03/2020 - 11/2020**

Principal investigator of the project: Search and characterization of star clusters in the far end of the Galactic bar | *CITEVA, Universidad de Antofagasta*

Development of a methodology based on unsupervised machine learning to search for young cluster candidates at the far end of the Galactic bar. This work allowed us to identify three new cluster candidates at a highly extincted and crowded region of the Milky Way. Under the supervision of Dr. Sebastián Ramírez and Dra. Karla Peña Ramírez.

Research Assistant **03/2020 - 11/2020**

Spectroscopy models for planetary atmospheres | *CITEVA, UA*

Study on transmission spectroscopy *in silico* models for Earth-like and Mini-Neptunes planetary atmospheres. The result of this research was used for a JWST Time Proposal (#2684, cycle 1). Under the supervision of Dr. Jeremy Tregloan-Reed.

Observer **2015 - 2025**
1M2H Project, UC Santa Cruz | *Las Campanas Observatory, Atacama, Chile*
Photometric transients follow-up using Swope Telescope. Responsible for data acquisition, data quality assurance, and system troubleshooting. Under the supervision of Dr. Ryan Foley.

Research Assistant **01/2016 - 06/2016**
Data reduction | *Instituto de Astronomía, Universidad Católica del Norte, Antofagasta, Chile*
Data reduction of VLT FORS and Magellan MagE data. Under the supervision of Dr. Christian Moni Bidin.

Observer **2014 - 2015**
Carnegie Supernova Project II | *Las Campanas Observatory, Atacama, Chile*
Observations of low-redshift supernovae using the Swope Telescope at Las Campanas Observatory. Under the supervision of Dr. Mark M. Phillips.

Research Assistant **2014 - 2015**
Sonification of medical images | *Instituto de Música, Pontificia Universidad Católica de Chile, Santiago, Chile*
Development of a sonification tool to assist radiologists in the detection of breast cancer. In collaboration with Dr. Rodrigo Cádiz and Dr. Patricio de la Cuadra.

EDUCATION

Doctorate in Astrophysics and Astroinformatics **2022 - (2025)**
Universidad de Antofagasta | *Antofagasta, Chile*
PhD candidate. Thesis: "Characterization of Sagittarius dwarf galaxy at low Galactic latitudes"

Master in Astronomy **2018 - 2020**
Universidad de Antofagasta | *Antofagasta, Chile*
Thesis: "Search and characterization of star clusters in the far end of the Galactic bar".

Licentiate in Astronomy **2008 - 2014**
Pontificia Universidad Católica de Chile | *Santiago, Chile*
Graduation activity: "A new astrometric software for the Swope Telescope", developed at Las Campanas Observatory.

SKILLS

Software Proficient in developing data analysis, reduction, and visualization using Python and its main associated modules (Astropy, Matplotlib, Numpy, Pandas, Plotly, PyTorch, Seaborn, SciPy). Astronomical software: IRAF, DS9, TOPCAT. Other technologies: Bash, Docker, Git, LaTeX, PostgreSQL, R. Github repo: https://github.com/jorgeanais

Languages
- Spanish (native)
- English (professional competence)

TEACHING EXPERIENCE

Instructor **2023 - Present**
Duoc UC, School of Informatics and Telecommunications | *Santiago, Chile*
Teaching Machine Learning and Deep Learning courses for computer engineering students.

ACHIEVEMENTS

Scholarships

- 2024-2026 Full Scholarship for advanced human capital in Chile, Agencia Nacional de Investigación y Desarrollo de Chile.
- 2022-2023 Academic Excellence Scholarship for PhD students, Universidad de Antofagasta.
- 2019 Full scholarship to attend La Serena School for Data Science, AURA Observatory, Chile.

PUBLICATIONS

Full list on ADS https://ui.adsabs.harvard.edu/public-libraries/veXcb7VPROqU0r03axnXkg